Elaborate Feature Decoupling for Weakly Supervised Fine-Grained Object Detection in Remote Sensing Images

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***Abstract*—Currently, low-resolution fine-grained remote sensing images (RSIs) greatly affect the performance of object detectors. Meanwhile, neither weakly supervised object detection (WSOD) nor fine-grained object detection (FGOD) methods can simultaneously solve the realistic problem of dependency on top-scoring proposals during the detector training faced by WSOD, and the imbalance between fine-grained classification and localization tasks faced by FGOD. To address these issues, this article proposes a novel, elaborate feature decouple network (EFDNet), which is one of the first end-to-end frameworks to perform weakly supervised FGOD (WSFGOD) in RSIs. Specifically, a lightweight multiorder degradation (LMD) module is introduced to better simulate complex real-world degradations, thus obtaining high-resolution image features by a modular connection method of multistage feature supplementation. Our adaptive contextual perception refinement (ACPR) module aims to adaptively shift the attention of the detection network from the local feature part to the whole object by integrating local and global contextual information. Finally, we propose a feature decoupled head (FDH) module to handle the fine-grained classification and localization tasks by the classification branch (CB) and localization branch (LB), respectively. Among FDH, CB provides rich semantic information for the classification task, while LB provides more detailed texture and edge information to delineate object boundaries accurately. Extensive experiments on the challenging FAIR1M-v1.0 and ShipRSImageNet datasets demonstrate that our proposed method achieves state-of-the-art performance and is highly effective in addressing multiscale object issues.**

***Index Terms*—Contextual information, feature decouple, super-resolution process, weakly supervised fine-grained object detection (WSFGOD).**

I. Introduction

With the development of modern remote sensor technologies, object detection from remote sensing images (RSIs) has emerged as an essential task for various aerial and satellite image applications. Traditional fully supervised object detection methods usually require fine bounding boxes to train models, but the labeling of bounding boxes is time-consuming and laborious, especially in RSIs. Moreover, these methods are also less scalable when dealing with fine-grained categories, where subtle differences between classes make annotation even more demanding. On the contrary, weakly supervised object detection (WSOD) has gained increasing attention due to its requirement for only image-level annotations. As an important research direction in the field of computer vision nowadays, typical WSOD approaches [1], [2], [3], [4] in RSIs are mainly inspired by the object detection algorithms developed for natural scenes and endeavor to address the suboptimal problem. Despite their successes, such ill-posed solutions ignore the property of RSIs. Different from natural images, remote sensing object detection still faces numerous difficulties and challenges, such as scale variations, small and dense objects, and complex backgrounds.

Most WSOD methods [1], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23] are designed for coarse-grained object detection and fail to capture the subtle differences between fine-grained categories. This limitation arises because they often rely on insufficient general features for distinguishing similar objects in RSIs. Fine-grained detection requires a holistic understanding of the image context to disambiguate similar objects. Many weakly supervised methods focus on local regions but neglect to integrate global contextual information, leading to suboptimal performance. In addition, for the fine-grained object detection (FGOD) field, high-definition RSIs are rich in semantic information, and the classification and localization tasks of object detection heavily rely on the semantic information encoded by advanced semantic-rich features. To improve the detection of objects at different scales, the feature pyramid network (FPN) [24] effectively distills the semantic information from high-level, semantic-rich features and integrates it into low-level feature maps that preserve fine details. This architectural design enhances the availability of semantic information for early-stage features and enables the detection of objects at various scales. Actually, bounding box regression typically requires detailed texture information and edge cues to accurately delineate object boundaries, while object classification benefits from richer semantic information. However, the fine-grained classification and localization tasks in RSIs pose new challenges. For instance, most existing methods demonstrate satisfactory performance on the DOTA [25] dataset but struggle to accurately

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identify fine-grained objects within the FAIR1M-v1.0 [26] dataset.

According to the above analysis, as shown in Fig. 1(a) and (b), neither WSOD nor FGOD can simultaneously address the following three challenges.

1) *Low-Quality Resolution of Image Features for Object Detectors*: Current traditional FGOD methods perform poorly on tiny object detection in low-resolution image features. The degradation of image quality directly impacts the ability to recognize subtle interclass variations, fundamentally exacerbating fine-grained confusion.
2) *Dependency on Top-Scoring Proposals*: These proposals on most WSOD methods often tend to cover only a portion of the object or even confused backgrounds, particularly in large-scale cluttered RSIs. Even when image resolution is inadequate, the initial proposal generation stage produces unreliable region proposals, creating a cascading effect that compromises the entire WSOD pipeline.
3) *Imbalance Between Fine-Grained Classification and Localization Tasks*: The classification task focuses on extracting effective semantic information to assign appropriate class labels to objects, while the localization task focuses on obtaining detailed boundary information to locate objects accurately. Furthermore, RSIs' characteristics of complex backgrounds and the susceptibility of objects to occlusion increase the difficulty of addressing the above three challenges. While previous works have addressed subsets of these challenges, no existing framework explicitly integrates solutions for all three challenges within an end-to-end architecture, especially when the vicious cycle caused by low-resolution features and poor proposals prevents both classification and localization tasks from performing optimally.

In this article, we propose an elaborate feature decouple network (EFDNet), which represents one of the first end-to-end frameworks explicitly designed to jointly address the challenges of weak supervision, fine-grained discrimination, and image degradation in RSIs. To better perform the weakly supervised FGOD (WSFGOD) task, our EFDNet consists of three modules, that is, lightweight multiorder degradation (LMD), adaptive contextual perception refinement (ACPR), and feature decoupled head (FDH).

For the problem of low-quality resolution of image features for object detectors, our LMD module adopts a multiorder degradation model for real-world degradations based on the classical one-order degradation model. It fully utilizes the low-frequency and high-frequency information by combining the features of each stage to maximally retain the information that is effective for high-resolution image features.

To tackle the problem of dependency on top-scoring proposals, we intricately introduce an end-to-end ACPR module. It aims at redirecting the attention of the detection network from local distinctive parts to the entire object and potential additional instances. This is achieved by harnessing both local and global context information.

To address the problem of imbalance between fine-grained classification and localization tasks, we meticulously devise a novel FDH module. For the fine-grained classification task, we propose a classification branch (CB) to extract fine-grained semantic features using fake attention maps for better classification accuracy. CB combines feature maps from different levels to create a rich feature map and generates multihead attention maps to weight features, improving attention quality. For the localization task, we introduce a localization branch (LB) that extracts high-resolution feature maps with edge information, merging detailed edge features with comprehensive object views to improve localization accuracy.

Based on the above analysis, the main contributions of this article are as follows.

1) We are the *first* to make studies on the WSFGOD field by proposing EFDNet. Our EFDNet simultaneously addresses the RSI's problems of dependency on top-scoring proposals existing in WSOD and the imbalance between classification and localization tasks existing in FGOD.
2) We present a novel LMD module to simulate practical degradations. It preserves essential low and high-frequency details for image feature resolution improvement by combining features from multiple stages.
3) We propose an ACPR module that utilizes contextual information from surrounding regions to suppress low-quality parts of objects and emphasize high-quality whole objects.
4) We design a novel FDH module to tackle the fine-grained classification and localization tasks. Among FDH, CB facilitates the learning of attention and fake attention feature maps, enhancing the quality of attention for fine-grained features while processing classification tasks, while LB extracts texture details and boundary information, which is beneficial to accurate localization tasks.

II. Related Works

A. Fine-Grained Object Detection

Compared with previous fine-grained recognition works based on the classification task [27], [28], [29], [30], FGOD requires simultaneous localization and fine-grained recognition. Since FAIR1M was served as a tailored benchmark for FGOD in RSIs, an increasing number of algorithms [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41] have been proposed since then. Current FGOD methods mainly make efforts to alleviate the semantic confusion between fine-grained classes. Zhou et al. [31] suggested attention-based group feature enhancement and subsaliency feature learning. Wang et al. [32] integrated an extra backbone to learn fine-grained classification and adopted knowledge distillation to keep it lightweight. By prototypical contrastive learning, PCLDet [42] maximized the interclass distance and minimized the intraclass distance. To better capture discriminative features, SFRNet [33] introduced a spatial and channel transformer and adopted metric learning to enhance the separability of fine-grained classes. Refined and balanced feature pyramid network (RB-FPN) [43] refined features extracted from the FPN using an attention module and a label assignment strategy, which dynamically modified the intersection over union (IoU) threshold according to statistical characteristics.

Besides, several approaches are designed to perform FGOD on the specific coarse-grained class. As for fine-grained

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ship detection, MGANet [34] built a self-attention network to exploit the global and local features. For fine-grained aircraft detection, instance switching-based contrastive learning (ISCL) [36] extracted various discriminative features by instance switching-based contrastive learning. Class-balanced contrastive learning (CBCL) [44] constructed class-balanced memory queues to address the class imbalance by memorizing training samples.

Instead of directly enhancing fine-grained recognition designed by the most recent FGOD methods, we apply super-resolution processing to proposal enhancement to overcome these obstacles caused by region proposals, which greatly enhances the fine-grained information of objects.

B. WSOD in RSIs

With the proliferation of weakly supervised generic object detection methods, WSOD in RSIs has gained much attention and evolved into a well-studied research field in these years.

In terms of WSOD methods based on multiple instance learning (MIL), dynamic course learning (DCL) [1] dynamically assigned easy-to-difficult labels to the training samples and iteratively trained detectors under the supervision to mine high-quality instances. Using a spatial graph-based voting mechanism and an appearance graph instance mining model, multiple instance graph (MIG) [2] detected all possible object instances with the same class and thus found high-quality objects.

In terms of methods based on contextual information, progressive contextual instance refinement (PCIR) [3] focused the model on the whole object and used a progressive object box self-pruning scheme to eliminate redundant candidate boxes, while TCANet [10] tackled the issue of grouped instances in RSIs by using a triple context-aware network, which fully utilized the global scene and internal context information of object instances.

In addition, self-supervised adversarial and equivariant network (SAENet) [11] introduced an adversarial dropout-activation block and a self-supervised transformation equivariance mechanism. By excavating more reliable candidate proposals, self-guided proposal generation (SGPG) [12] explicitly generated more high-quality instance-level features. To facilitate rotation-invariant learning, RINet [4] detected its predictions for prerotation and postrotation instances through multibranch online detector optimization. Pseudo-instance soft labels (PSLs) [15] solved part domination and misclassification induced by copying and neighbor instances' pseudo-labels. Weakly supervised-oriented detector (WSODet) [14] achieved a prediction-oriented bounding box with only image-level labels. Self-training-based semantic-balanced network (STSNet) [17] and PILFR [18] networks solve missing detection and partial domination problems in an end-to-end manner.

Different from the above methods, our approach keeps the balance between contextual information for the grouping instance problem and multiple feature extraction for the multiscale problem in RSIs.

III. Methodology

A. Overview of the Proposed Method

The overview of the proposed EFDNet is outlined in Fig. 2. The LMD first fuses the feature maps extracted from the shallow stage of the model with those extracted from the deep stage of the model, to facilitate the obtaining of higher-resolution image features. In the WSOD framework, the ACPR module adaptively then extracts precise bounding boxes by integrating contextual information from surrounding regions, while identifying instances in images by assessing the contextual similarity between the top-scoring proposals and other proposals. Among the FDH module, the CB effectively extracts fine-grained semantic information by introducing fake attention maps for the fine-grained classification task. For the localization task, the LB extracts high-resolution feature maps enriched with additional edge information.

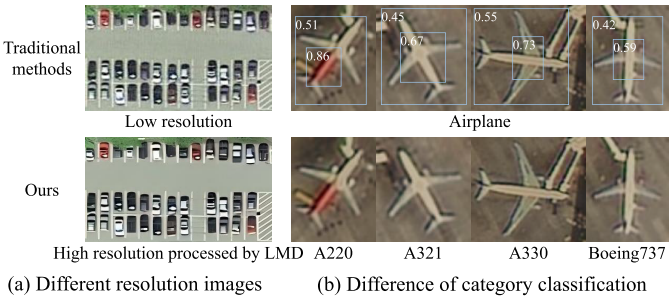


Fig. 1. Difficulties and phenomena of WSFGOD in RSIs. (a) Influence of low-resolution image features on the performance of fine-grained object detectors. (b) Comparison of the classification and localization tasks between ordinary and fine-grained remote sensing object detection; selecting only the most confident and top-scoring proposals to train the detector (first row).

B. Lightweight Multiorder Degradation

When adopting the classical degradation model to synthesize training pairs, the model developed is capable of managing certain real samples [45], [46], [47], [48], [49]. Among them, BSRGAN [45] proposed a degradation model that included random confusion blur, downsampling, and noise, while Real-ESRGAN [46] is engineered to more accurately mimic intricate real-world degradation through the incorporation of a higher-order degradation modeling process. Moreover, Zhang et al. [48] focused on blind deblurring and developed a pixel screening-based intermediate correction method. However, these off-the-shelf methods remain incapable of addressing complex degradations encountered in the real world, particularly those involving unknown noises, which is because the synthetic low-resolution images still exhibit a significant disparity from real-world degraded images. The key rationale for integrating degradation simulation within the network lies in the end-to-end optimization and feature-level adaptation, rather than simply using the above super-resolution preprocessing step.

Thus, we propose an LMD module extended by the classical degradation model to simulate more practical degradations. Unlike the above existing methods, limited to synthetic noise and blur, LMD advances degradation modeling by extending classical single-order degradation frameworks to a multiorder process. An n-order model involves n repeated degradation processes p

$$p = D^n(q) = (D_n \circ \cdots \circ D_2 \circ D_1)(q) \tag{1}$$

where the degraded low-resolution image p is obtained by adding noise in the classical degradation model, q represents

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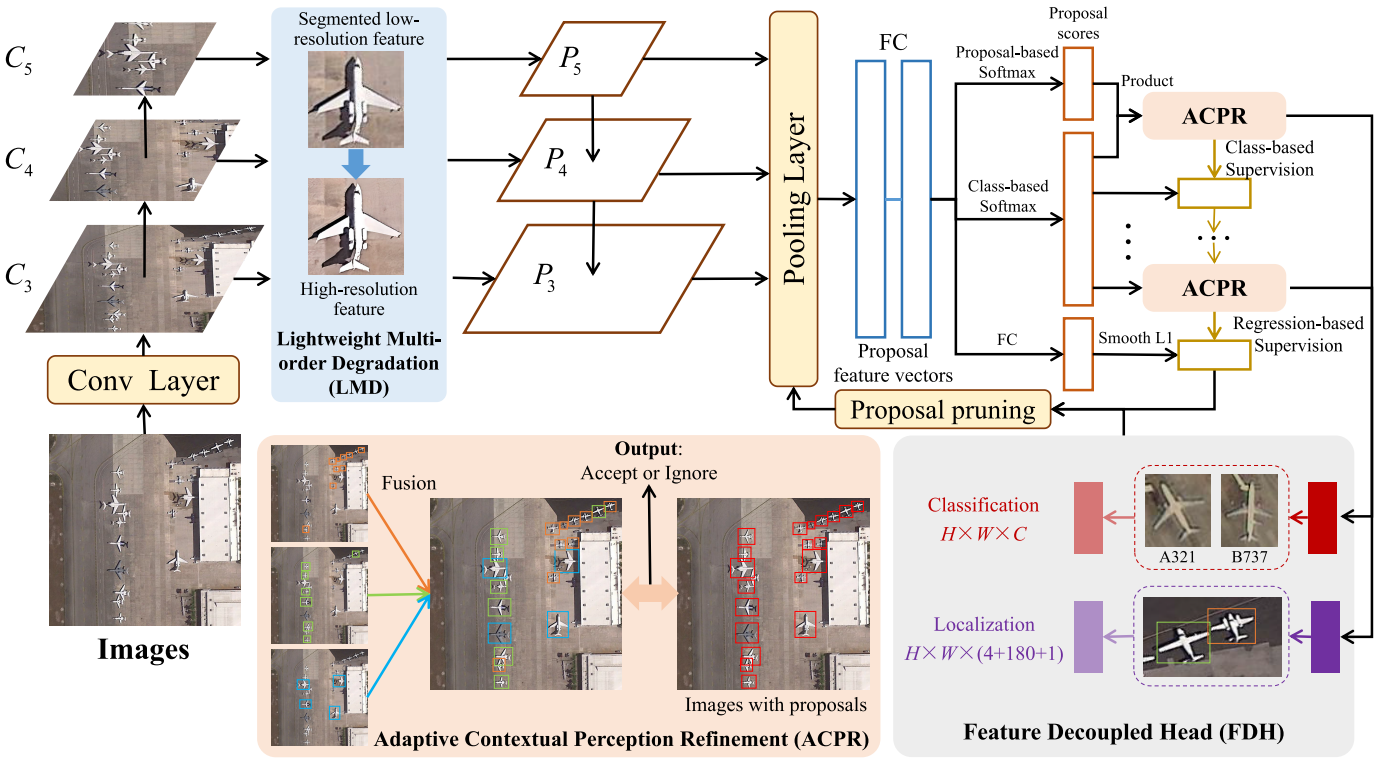


Fig. 2. Illustration of the proposed EFDNet. EFDNet is mainly composed of the LMD module, the ACPR module, and the FDH module.

TABLE I

RESULTS (%) FOR DIFFERENT COMPONENTS OF EFDNET ON THE FAIR1M-V1.0 AND SHIPRSIMAGENET TESTING/TRAIN–VAL DATASETS

Modules			Inference	FLOPs	FAIR1M-v1.0					ShipRSImageNet				
LMD	ACPR	FDH	Time (ms)	(G)	AP_S	AP_M	AP_L	mAP	CorLoc	AP_S	AP_M	AP_L	mAP	CorLoc
Baseline PCIR [3]			22.4	147.43	15.90	31.30	46.55	31.69	59.77	22.17	53.50	67.03	47.57	67.31
✓			24.9	140.54	20.14	34.08	47.70	33.97	62.88	23.31	56.27	70.51	50.03	70.83
	✓		25.0	140.15	21.85	33.10	46.48	33.81	62.03	24.91	54.55	68.60	49.35	69.87
		✓	24.5	140.93	20.10	34.01	47.59	33.90	62.74	23.27	56.14	70.34	49.91	70.66
✓	✓		25.4	134.86	21.99	35.27	49.42	35.56	65.55	25.27	58.20	73.05	52.17	73.84
✓		✓	25.1	135.64	21.13	35.73	50.01	35.62	65.92	24.45	59.00	73.92	52.46	74.25
	✓	✓	24.7	136.23	21.89	35.09	49.19	35.39	65.22	25.14	57.89	72.66	51.90	73.46
✓	✓	✓	25.6	132.02	**22.01**	**37.23**	**52.12**	**37.12**	**68.86**	**25.67**	**62.91**	**79.54**	**56.04**	**77.96**

TABLE II

RESULTS (%) FOR THE ABLATION STUDY OF THE LOSS FUNCTION ON THE FAIR1M-V1.0 AND SHIPRSIMAGENET DATASETS

Methods / Loss Function			Inference	Parameters	FLOPs	FAIR1M-v1.0		ShipRSImageNet	
$L(Y,S_c)$	L_r^k	L_{reg}	Time (ms)	(M)	(G)	mAP	CorLoc	mAP	CorLoc
Baseline PCIR [3]			22.4	138	147.43	31.69	59.77	47.57	67.31
✓			25.1	195	143.44	32.53	60.35	49.11	68.32
✓	✓		25.5	195	137.67	34.97	64.87	52.79	73.44
✓	✓	✓	25.6	195	132.02	**37.12**	**68.86**	**56.04**	**77.96**

the ground-truth high-resolution image, D_i denotes the distinct degradation operation, and $(D_n \circ \cdots \circ D_2 \circ D_1)(q) = (D_n(\cdots(D_2(D_1))))(q)$. Specifically, each D_i can be defined as

$$D_i = B_i \circ R_i \circ N_i \circ C_i \tag{2}$$

where each degradation operation step D_i is explicitly defined as a composite function that applies a convolution blur B_i, stochastic resizing R_i using bilinear interpolation, additive Gaussian noise N_i, and image compression C_i. Crucially, our analysis emphasizes that only a subset of the theoretically possible shuffled degradations is indispensable, signifying that the multiorder aspect itself is fundamental, not the application of every permutation. To maintain image resolution within a practical range, a stochastic resizing operation substitutes conventional downsampling. We utilize a second-order degradation paradigm, balancing fidelity to diverse real-world scenarios with computational efficiency, which is proved in LMD's multiorder design of section experiments.

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TABLE III

RESULTS (%) FOR THE ABLATION STUDY OF THE MULTIORDER DESIGN IN LMD ON THE FAIR1M-V1.0 AND SHIPRSIMAGENET DATASETS. (HERE, THE BEST AND SECOND-BEST MAP/CORLOC ARE, RESPECTIVELY, HIGHLIGHTED IN RED AND BLUE)

Methods	Backbone	LMD	Inference Time (ms)	Parameters (M)	FLOPs (G)	FAIR1M-v1.0 mAP	FAIR1M-v1.0 CorLoc	ShipRSImageNet mAP	ShipRSImageNet CorLoc
Baseline PCIR [3]	VGG16	/	22.4	138	147.43	31.69	59.77	47.57	67.31
Baseline PCIR [3]	R-50-FPN	/	48.1	25.6	174.58	32.18	60.70	48.30	68.35
Ours	VGG16	1-order	25.8	145	142.07	33.35	61.87	50.35	70.04
Ours	VGG16	2-order	25.6	195	132.02	37.12	68.86	56.04	77.96
Ours	VGG16	3-order	25.5	268	134.56	37.84	70.20	57.13	79.47
Ours	R-50-FPN	1-order	55.5	30	167.67	34.06	63.18	51.42	71.53
Ours	R-50-FPN	2-order	55.1	42	164.47	37.61	69.77	56.78	78.99
Ours	R-50-FPN	3-order	55.3	56	162.73	38.34	71.12	57.88	80.52
Ours	LSKNet	1-order	62.1	88.6	305.63	35.38	65.63	53.41	74.31
Ours	PKINet	1-order	66.2	91.2	355.36	35.76	66.34	53.99	75.10
ORCNN	VGG16	/	24.1	155	13.60	36.70	68.08	55.41	77.08

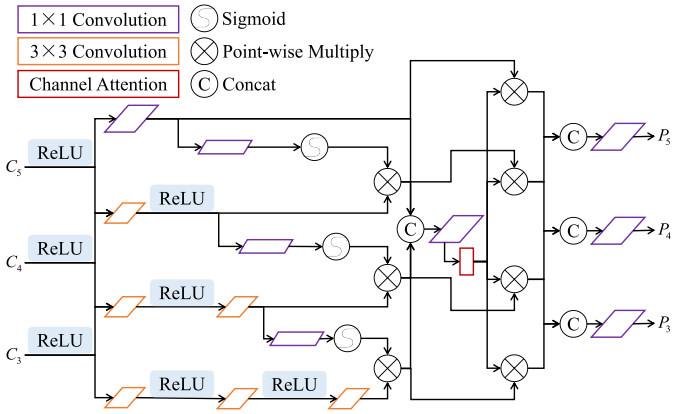


Fig. 3. Illustration of the LMD.

Fig. 3 depicts the internal architecture of the LMD module. LMD improves feature representation robustness by incorporating intermediate feature outputs from the preceding four processing stages. These features undergo deep fusion and restructuring within LMD, facilitating comprehensive integration between shallow and deep semantic layers. This strategy is designed to mitigate information loss throughout training, avoiding the domain gap between preprocessing and the main task. Specifically, LMD's lightweight design utilizes a four-branch progressive fusion framework to methodically propagate and combine shallow features with subsequent, deeper abstractions. This structured hierarchical blending maximizes the effective utilization of informative detail present in shallow maps and contextual knowledge embedded within deeper layers, thereby optimizing image resolution enhancement efficiency by sharing features across stages while minimizing the degradation of feature resolution fidelity.

C. Adaptive Contextual Perception Refinement

As the baseline of our network, PCIR [3] is referred to as our basic framework for WSOD because of its scalability and effectiveness. Different from PCIR [3], we propose the ACPR module that focuses the model on the whole object and uses a progressive object box self-pruning scheme to eliminate redundant proposal boxes. Fig. 2 shows that it consists of several refinement subbranches, each responsible for refining object proposals by mapping their feature vectors into a multiorder fractional vector. Each branch focuses on different refinements, which improve the accuracy of the proposed bounding boxes.

Given an input image $I^{(i)}$ and its proposals R. In the upper part of the basic WSOD framework, let $Y = [Y_1, \ldots, Y_c, \ldots, Y_C] \in \{0, 1\}$ denote the binary image-level label to indicate the presence or absence of the specific class objects in this image. Following two fully connected (FC) layers, the feature vector of each proposal is split into two streams: the upper stream categorizes each proposal into different classification containers, while the lower one is dedicated to detection. The score of each individual proposal x_{cr} is generated from the elements between the results of these two streams, denoted by the special symbol $\odot$, i.e.,

$$x_{cr} = \frac{e^{x_{ij}^c}}{\sum_{k=1}^{N_C} e^{x_{kj}^c}} \odot \frac{e^{x_{ij}^d}}{\sum_{r=1}^{N_R} e^{x_{ir}^d}} \tag{3}$$

where x_{ij}^c and x_{ij}^d, respectively, refer to feature vectors in the classification and detection stream for each individual proposal, the special symbol $\odot$ represents the elemental corresponding multiplication by two streams, N_C and N_R, respectively, represent the number of image categories and proposals. Then, by summing each proposal' score x_{cr}, the image-level score S_c for the cth class is calculated as

$$S_c = \sum_{r=1}^{N_R} x_{cr}. \tag{4}$$

Drawn on the multiclass cross-entropy loss function, the loss function of the two-stream network is formulated as

$$L(Y, S_c) = -\sum_{c=1}^{N_C} \{Y \log S_c + (1 - Y) \log (1 - S_c)\}. \tag{5}$$

Along with our two-stream network of the WSOD framework, several ACPR branches are introduced. Each ACPR refinement branch maps the eigenvector of the proposal to a $\{N_C + 1\}$-dimensional fractional vector $x_{cr}^k \in \mathbb{R}^{(N_C+1)\times 1}, k \in (1, \ldots, K)$, where k represents the kth refinement branch and $\{N_C + 1\}$ represents N_C different object categories and a background category.

In each ACPR branch, the WSOD framework utilizes the local context information to ascertain the most representative

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holistic object and further leverages global context information to discover other possible instances present in the image. Denote $y_{cr}^k \in (0, \ldots, N_C)$ as the pseudo-label for the rth proposal in the kth refinement branch, and the loss function for the kth refinement is defined as

$$L_r^k = -\frac{1}{N_R} \sum_{r=1}^{N_R} \sum_{c=1}^{N_C+1} y_{cr}^k \log x_{cr}^k. \tag{6}$$

After a series of operations involving ACPR, pooling, and full connectivity, a regression branch is placed at the bottom of the basic WSOD framework. To enable end-to-end training of the weakly supervised detection model and simplify the training process, we construct a single end-to-end network by directly adding the regression branch to our WSOD framework. Meanwhile, in order to more accurately locate object information in the feature map, this regression branch introduces regression supervision, so as to predict the coordinates of each object frame while dynamically correcting the position information of the object frames by applying weak supervision.

Let x_{reg} represent the proposal's score in the regression branch, so the predicted values of the regression branch model can be expressed as $z(x_{\text{reg}}) = \sigma(x_{\text{reg}})$, where $\sigma(\cdot)$ is the regression function. Then, for the whole model training process of N images, the sample true values are $y = \{y^{(1)}, y^{(2)}, \ldots, y^{(N)}\}$, and the predicted values of the object detection model are $\{z(x_{\text{reg}})\} = \{z(x_{\text{reg}}^{(1)}), z(x_{\text{reg}}^{(2)}), \ldots, z(x_{\text{reg}}^{(N)})\}$. Consequently, we derive the corresponding Smooth L1 loss function L_{reg}. Therefore, we ultimately train the detection network by combining the loss functions of $L(Y, S_c)$, L_r^k, and L_{reg}, obtaining the total loss function as

$$L = L(Y, S_c) + \sum_{k=1}^{K} L_r^k + L_{\text{reg}} \tag{7}$$

where K represents the total number of refinement times.

With the optimization of the total loss, the object detector can more accurately and dynamically correct the proposals during the training process, while PCIR [3] prunes them only on classification scores and geometric compatibility. Specifically, for proposal pruning, it first generates proposals based on an initial set of object candidate regions, followed by the two-stage WSOD framework. Then these proposals are pruned based on their confidence scores and geometric compatibility with the ground truth, thereby improving the detection performance of tiny objects.

To avoid two adjacent objects in the PICR's mechanism being mistakenly merged into one, ACPR adopts a multistage contextual reasoning process, leveraging both local spatial relationships and global semantic consistency. Specifically, it leverages contextual dissimilarity between merged regions and their surroundings and utilizes regression-based self-correction to shrink oversized proposals.

D. Feature Decoupled Head

Current object detection methods often use a coarse-grained approach, focusing on key object features known as salient features. However, for fine-grained subcategory classification, this method falls short due to similarities between different subcategories and differences within the same subcategory.

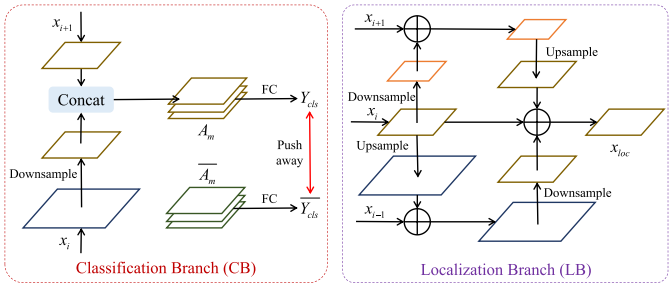


Fig. 4. Illustration of FDH.

To accurately classify fine-grained categories, more discriminative semantic features are needed. In contrast, localization requires detailed texture and boundary information, with lower pyramid levels of feature maps providing a stronger response to object contours, edges, and textures.

Therefore, we propose the FDH for efficient and accurate fine-grained classification and localization. In order to combine the high-dimensional information of the deep feature map with the detailed information of the shallow feature map, we propose the CB and LB to extract scale-invariant features and achieve precise localization without compromising efficiency. Different from YOLOX's decoupled head [50] focuses on decoupling the two tasks to increase efficiency, our FDH introduces a more flexible mechanism that integrates fake attention maps and a dedicated LB, thereby adapting to multiscale feature fusion and incorporates FGOD. YOLOX [50] uses standard convolutional heads for task decoupling but does not address the subtle interclass variations in fine-grained scenarios. In contrast, CB of our FDH employs fake attention maps to regularize attention distribution, preventing overfitting to dominant features. Simultaneously, LB leverages edge-enhanced features from multiple pyramid levels, which are uniquely suited for remote sensing objects with complex boundaries and scale variations.

Specifically, at each pyramid level i, CB leverages the feature maps from two levels, x_i and x_{i+1}, to generate a semantically rich feature map for classification. As shown in Fig. 4, we downsample x_i by a factor of 2 and concatenate it with x_{i+1} to generate the final x^{cls}

$$x_i^{\text{cls}} = \text{Down}(x_i) \oplus x_{i+1} \tag{8}$$

where $\oplus$ and $\text{Down}(\cdot)$ represent concatenation and a downsampling convolutional layer. This structure can encode more semantic information.

The resulting feature maps x^{cls} are denoted as $x^{\text{cls}} \in \mathbb{R}^{C*H*W}$, where H, W, and C represent height, width, and the number of channels. Subsequently, M multihead attention maps are generated by applying a simple convolutional function to the extracted features x^{cls}. These attention maps are represented as $A = \phi(x^{\text{cls}})$, where $\phi(\cdot)$ signifies the convolutional function with a kernel size of 1×1 and the number of attention maps M. The set of attention maps generated is denoted as $A = \{A_1, A_2, \ldots, A_M\}$. Notably, $A_M \in A$ represents the attention maps that highlight interesting regions in the input image. The feature maps x^{cls} are then weighted by the attention maps to generate fine-grained features $G_m = \phi(x^{\text{cls}} \cdot A_m)$ after global average pooling, where ϕ represents the global average pooling

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operation, and $\cdot$ denotes element-wise multiplication. The final prediction $Y_{x^{\text{cls}}}(A)$ is obtained by applying a classifier to the sum of the weighted features

$$Y_{x^{\text{cls}}}(A) = fc\left(\sum_{m=1}^{M} G_m\right) = fc\left(\sum_{m=1}^{M} \phi\left(x^{\text{cls}} \cdot A_m\right)\right) \tag{9}$$

where $fc(\cdot)$ represents a classifier including a fully convolutional layer and a batch normalization layer. To enhance attention quality for fine-grained features, we introduce the concept of fake attention maps. The fake attention map is an attention map simply generated by a random function. Different from deterministic attention that may bias models toward salient regions, a random sampling function for fake attention maps acts as a form of attention regularization, preventing reliance on object-centered patterns and reducing spatial bias. Specifically, fake attention maps act as a regularizer by adding stochasticity during training, similar to adversarial training paradigms, which improves generalization and reduces overfitting by ensuring the model focuses on subtle, discriminative features across diverse regions of the image rather than memorizing limited parts. This flexibility and lack of structure enable the model to freely learn robust and unbiased features for fine-grained classification, unlike more rigid strategies such as uniform sampling, Gaussian sampling, random Gaussian noise, or spatially sparsified sampling, which could inadvertently limit the model's learning capacity. Therefore, the fake prediction $Y_{x^{\text{cls}}}(\overline{A})$ is expressed as

$$Y_{x^{\text{cls}}}\left(\overline{A}\right) = fc\left(\sum_{m=1}^{M} \overline{G_m}\right) = fc\left(\sum_{m=1}^{M} \phi\left(x^{\text{cls}} \cdot \overline{A_m}\right)\right) \tag{10}$$

where $\overline{A}$ represents fake attention maps. The randomly generated fake attention maps are designed to be independent of object positions, avoiding bias toward object centers or predefined areas. By introducing these maps, the model explores diverse feature areas, helping to avoid overfitting and improve generalization. This process encourages the network to focus on generalized, nonsalient features, making it more robust to variations in object appearance. The sparse, random distribution of fake attention maps ensures they do not excessively overlap with the objects, promoting a more balanced feature learning process.

CB of FDH implements a dual-attention learning paradigm. This framework jointly optimizes authentic attention mappings and synthetically generated counterfeit counterparts, significantly refining discriminative focus on subtle fine-grained characteristics. Crucially, the counterfeit branch proportionally down-weights activation in high-saliency regions, concurrently suppressing distraction from irrelevant background elements. Our methodology inherently compels attention mechanisms to holistically evaluate both salient and suboptimal feature signatures through explicit comparison of learned versus artificial attention distributions. The final prediction Y_{cls} is obtained by taking the difference between the attention prediction $Y_{x^{\text{cls}}}(A)$ and its fake prediction $Y_{x^{\text{cls}}}(\overline{A})$

$$Y_{\text{cls}} = Y_{x^{\text{cls}}}(A) - Y_{x^{\text{cls}}}\left(\overline{A}\right). \tag{11}$$

Feature hierarchies inherently encode scale-dependent positional semantics: higher-resolution feature maps encapsulate low-dimensional textural details essential for localizing smaller objects, while condensed representations retain high-dimensional abstractions for larger objects. To address this multiscale challenge, FPN [24] introduces a structured cross-resolution fusion architecture. Its bidirectional cross-connections enable efficient interscale feature exchange, substantially enhancing detection robustness across diverse object scales. While FPN [24] is a general method for multiscale feature extraction, CB is specifically designed to enhance the localization effect by combining detailed edge features with the global context. Furthermore, CB focuses on localization and fuses edge features from lower pyramid levels with semantically rich global features from deeper layers through upsampling and downsampling.

As for LB of FDH, feature maps are taken from levels x_{i-1}, x_i, and x_{i+1} at each pyramid level i. x_{i-1} contributes more detailed and edge features, while x_{i+1} provides a more comprehensive perspective of objects. The structure of LB is depicted in Fig. 4. Initially, x_i is upsampled by a factor of 2 and aggregated with x_{i-1}. Subsequently, a 3×3 convolutional layer with a stride of 2 downsamples to match the resolution of x_i. Following this, x_i is downsampled, aggregated with x_{i+1}, and then upsampled and aggregated to generate the final x^{loc}. The computation is expressed as

$$\begin{aligned} x_i^{\text{loc}} = x_i &+ \text{Up}\left(x_{i+1} \oplus \text{Down}\left(x_i\right)\right) \\ &+ \text{Down}\left(x_{i-1} \oplus \text{Up}\left(x_i\right)\right) \end{aligned} \tag{12}$$

where $\text{Up}(\cdot)$ represents the upsampling convolutional layer.

IV. Experiments

A. Datasets and Evaluation Metric

Our network is evaluated on the commonly used FAIR1M-v1.0 [26] and ShipRSImageNet [55] datasets. FAIR1M is a large-scale and tailored benchmark for FGOD and recognition in RSIs, with most experiments in the field utilizing either version 1.0 or 2.0. FAIR1M-v1.0 [26] includes five object coarse categories, 37 subcategories and 42 796 images, comprising more than 1 million instances annotated by oriented bounding boxes (OBB). The image resolutions range from 1000×1000 to $10\,000 \times 10\,000$. ShipRSImageNet [55] is a well-known public dataset for fine-grained ship detection in high-resolution optical RSIs. ShipRSImageNet [55] includes 50 different categories and 3435 images at a resolution of 930×930 pixels, containing 17 573 ship instances from various sensors, satellite platforms, locations, and seasons.

Following the standard routine in WSOD, our network is trained on both the training and validation (referred to as the train–val) sets, and evaluated on the testing set. Meanwhile, only image-level labels are available during the model training. Mean average precision (mAP) and correct localization (CorLoc) accuracy are employed to evaluate the accuracy of object detection and localization, respectively. Inference time, parameters (Para.), and floating-point operations per second (FLOPs) are applied to measure the computational complexity and efficiency of the model. All metrics are performed on the PASCAL criteria (IoU = 50%).

B. Implementation Details

In terms of initialization, VGG16 [56] and ResNet-50 pretrained on the ImageNet [57] are mainly adopted as the

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backbone. All newly added layers are initialized with a Gaussian distribution with a 0-mean and 0.01-standard deviation. To protect the features of small-sized objects, we use dilated convolutional layers to replace the fourth max-pooling layer (pooling4) and its subsequent convolutional layers. Before training the network, our EFDNet generates approximately 2000 proposals for each image. We augment the training data with three rotation transformations {90°, 180°, 270°}. For horizontal bounding boxes, we rely on traditional methods that follow standard approaches for WSOD. For rotated bounding boxes, we employ rotation-invariant techniques that can handle various angle orientations in object localization.

During training, the network performs 30K and 300K iterations. Meanwhile, we keep the training settings, including learning rate, mini-batch, weight decay, and momentum, consistent with those in [6], [7], and [8]. They are set to 0.001, 2, 0.005, and 0.9, respectively. A confidence threshold of 0.3 is applied for nonmaximum suppression (NMS) [58] to eliminate duplicated bounding boxes.

C. Ablation Studies

In this section, we perform ablation experiments on both the FAIR1M-v1.0 [26] and ShipRSImageNet [55] datasets to reveal the contributions of the proposed LMD, ACPR, and FDH. To demonstrate the effectiveness of our module, we adopt a training strategy similar to that of PCIR [3], which selects only the highest proposal to refine the next classifier.

As presented in Table I, taking the FAIR1M-v1.0 [26] dataset as an example, the integration of LMD, ACPR, and FDH into our EFDNet alone results in respective increases in mAP of 2.28%, 2.12%, and 2.21%, along with improvements in CorLoc of 3.11%, 2.23%, and 2.97%, respectively, which indicates that all the modules have positive effects on performance enhancement. For LMD, the performance gain stems from its ability to simulate real-world degradations in an end-to-end manner, whereas off-the-shelf super-resolution preprocessing would introduce artifacts that misalign with the detection task. For FDH, the gain is attributed to its fine-grained-specific decoupling. FDH's fake attention maps and edge-aware LB reduce overfitting on fine-grained categories by enforcing attention diversity. When selecting any two of LMD, ACPR, and FDH modules for integration into our network, mAP, respectively, increases by 3.87%, 3.93%, and 3.70%, and CorLoc, respectively, improves by 5.78%, 6.15%, and 5.45%, which indicates that the improved accuracy of mAP and CorLoc is basically higher than the sum of the corresponding individual modules. The synergistic effect of combining modules such as LMD and FDH indicates that LMD enhances feature resolution, while FDH leverages these features for precise discrimination. As for the ShipRSImageNet [55] dataset, the integration of LMD, ACPR, and FDH into our EFDNet alone yields respective increases in mAP of 2.46%, 1.79%, and 2.35%, along with improvements in CorLoc of 3.52%, 2.55%, and 3.34%, while integrating any two of these modules into our network results in respective increases in mAP of 4.61%, 4.89%, and 4.33%, and respective improvements in CorLoc of 6.52%, 6.93%, and 6.14%. It is obvious that the LMD, ACPR, and FDH modules exhibit similar performance effects on the ShipRSImageNet [55] dataset as previously analyzed.

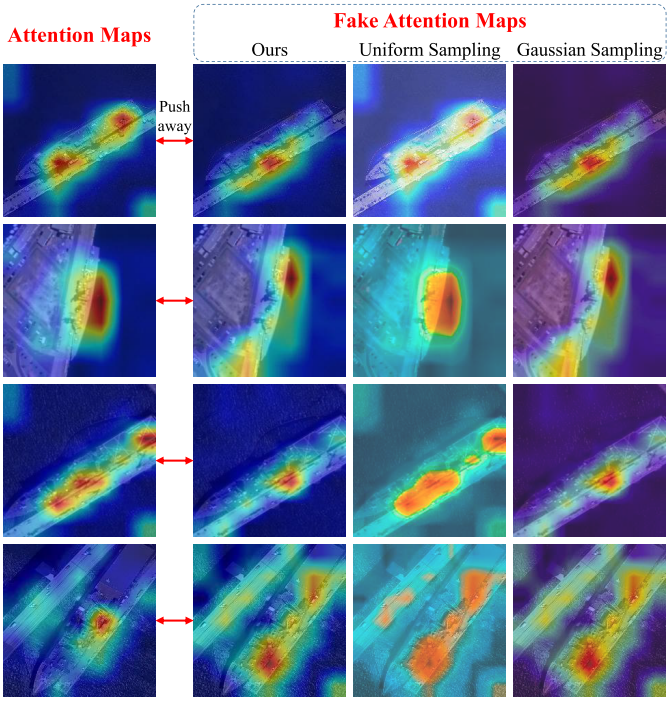


Fig. 5. Comparative ablation experiments of uniform sampling, Gaussian sampling, and our EFDNet on attention maps and fake attention maps.

As for computational cost of inference time and FLOPs, the baseline model PCIR [3] exhibits an inference time of 22.4 ms and FLOPs of 147.43G. The integration of LMD, ACPR, and FDH into our EFDNet alone results in respective increases in inference time of 2.5, 2.6, and 2.1 ms, along with decreases in FLOPs of 6.89G, 7.28G, and 6.50G, indicating that our modules achieve a significant improvement in mAP and CorLoc at the cost of a slight increase in inference time and decrease in FLOPs. For combined modules, the inference time and FLOPs remain manageable, which demonstrates that EFDNet achieves a favorable tradeoff between accuracy and efficiency, as the modules introduce minimal latency while significantly improving performance through computational optimization.

For multiscale issues in RSIs, our ACPR module improves the mAP metrics while improving the average precision of small objects (AP_S) compared to all SOTA methods. Here, AP_S, AP_M, and AP_L, respectively, denote small, medium, and large objects (they are divided by 32×32 and 96×96 pixels). Take the ablation experiment of the ACPR module presented in Table I as an example, it, respectively, reaches AP_S exceeding 21.5% and 24.9% on the FAIR1M-v1.0 [26] and ShipRSImageNet [55] datasets, demonstrating that ACPR effectively enhances the performance of small-object detection.

1) Fake Attention Maps: As depicted in Fig. 5, FDH architecturally enables simultaneous learning of complementary attention feature maps and their counterfeit counterparts. This dual-path paradigm progressively enhances attention quality for fine-grained discriminative attributes through adversarial optimization. Concurrently, the fake attention branch systematically attenuates focus on high-saliency regions, inducing a proportional redistribution of attention resources toward subtle class-specific characteristics. Moreover, random sampling

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TABLE IV
AVERAGE PRECISION (%) RESULTS FOR DIFFERENT WSOD METHODS ON THE FAIR1M-V1.0 TESTING DATASET

WSOD Methods	B737	B747	B777	B787	C919	A220	A321	A330	A350	ARJ21	OA	PS	MB	FB	TB	ES	LCS	DCS	WS
WSDDN (2016) [5]	48.62	40.38	26.85	28.11	21.04	48.23	52.03	34.92	23.47	24.46	49.26	5.40	27.60	15.02	29.73	28.91	35.46	38.47	5.71
OICR (2017) [6]	52.41	43.53	28.95	30.30	22.68	51.99	56.09	37.64	25.30	26.37	53.10	5.82	29.75	16.19	32.05	31.17	38.23	41.47	6.15
PCL (2018) [7]	53.70	44.59	29.66	31.04	23.24	53.26	57.46	38.57	25.92	27.02	54.40	5.96	30.48	16.59	32.84	31.93	39.16	42.49	6.30
PCIR (2020) [3]	55.09	45.75	30.42	31.84	23.84	54.64	58.95	39.57	26.59	27.72	55.81	6.12	31.27	17.02	33.69	32.76	40.18	43.59	6.47
DCL (2021) [1]	53.59	44.51	29.60	30.98	23.20	53.16	57.35	38.49	25.87	26.96	54.29	5.95	30.42	16.56	32.77	31.87	39.09	42.40	6.29
TCANet (2021) [10]	57.01	47.35	31.49	32.96	24.68	56.55	61.01	40.95	27.52	28.69	57.76	6.33	32.37	17.61	34.87	33.91	41.58	45.11	6.69
SAENet (2022) [11]	59.53	49.44	32.88	34.41	25.77	59.05	63.70	42.76	28.73	29.95	60.31	6.61	33.79	18.39	36.40	35.40	43.42	47.10	6.99
RINet (2022) [4]	61.52	51.92	33.53	36.14	27.06	62.01	66.90	44.90	29.18	31.46	63.34	6.94	35.49	20.31	38.23	36.18	45.60	49.47	7.34
WSONet (2023) [14]	60.49	50.24	33.41	34.97	26.18	60.00	64.73	43.45	29.20	30.44	61.28	6.72	34.34	18.69	36.99	35.97	44.12	47.86	7.10
PSL (2023) [15]	64.25	50.53	34.93	35.56	28.38	62.74	66.69	45.43	30.53	30.82	64.08	7.02	37.91	19.54	38.68	37.62	46.13	50.05	7.43
HIMDLA (2024) [16]	59.19	49.16	32.69	33.22	25.62	58.71	63.35	42.52	28.57	28.78	58.97	6.57	33.60	18.29	36.20	35.20	43.17	45.84	7.95
STSNet (2024) [17]	62.95	52.28	34.77	35.39	27.25	62.44	67.37	44.22	30.39	30.67	63.78	6.99	35.74	19.45	38.50	37.44	45.91	48.81	7.39
PILFR (2024) [18]	61.43	51.02	33.93	35.51	26.59	60.93	65.74	44.12	29.65	30.91	62.23	6.82	34.87	18.98	37.56	38.53	46.80	48.61	7.21
Ours	63.02	52.09	37.64	39.30	26.93	65.00	69.05	45.35	30.15	33.47	67.37	6.16	36.63	18.93	38.46	38.37	48.06	50.06	6.58

WSOD Methods	OS	SC	BUS	CT	DT	VAN	TRAI	TRAC	EX	TT	OV	BC	TC	FF	BF	IS	RO	BR	mAP
WSDDN (2016) [5]	4.70	41.20	21.94	31.15	27.95	40.37	2.44	1.09	6.42	16.55	1.05	38.17	55.00	43.68	52.35	35.17	17.17	14.80	**27.97**
OICR (2017) [6]	5.07	44.41	23.65	33.58	30.13	43.52	2.63	1.18	6.92	17.84	1.13	41.14	59.29	47.08	56.43	37.91	18.51	15.95	**30.15**
PCL (2018) [7]	5.19	45.50	24.24	34.41	30.86	44.58	2.70	1.20	7.09	18.28	1.16	42.15	60.75	48.24	57.81	38.84	18.96	16.34	**30.89**
PCIR (2020) [3]	5.33	46.68	24.86	35.30	31.66	45.74	2.77	1.24	7.28	18.75	1.19	43.25	62.32	49.49	59.31	39.85	19.46	16.77	**31.69**
DCL (2021) [1]	5.18	45.41	24.19	34.34	30.80	44.50	2.69	1.20	7.08	18.24	1.16	42.07	60.63	48.14	57.70	38.77	18.93	16.31	**26.83**
TCANet (2021) [10]	5.51	48.32	25.73	36.53	32.77	47.34	2.86	1.28	7.53	19.41	1.23	44.76	64.50	51.22	61.39	41.25	20.14	17.35	**32.80**
SAENet (2022) [11]	5.76	50.44	26.87	38.14	34.22	49.43	2.99	1.34	7.86	20.27	1.29	46.73	67.34	53.48	64.09	43.06	21.02	18.12	**34.25**
RINet (2022) [4]	6.04	52.98	28.22	40.06	35.94	51.91	3.14	1.40	8.26	21.28	1.35	49.08	72.73	57.16	67.31	45.23	22.08	19.03	**35.96**
WSONet (2023) [14]	5.85	51.26	27.30	38.76	34.77	50.23	3.04	1.36	7.99	20.59	1.31	47.49	68.43	54.34	65.13	43.76	21.36	18.41	**34.80**
PSL (2023) [15]	6.12	53.60	28.55	40.53	37.36	52.52	3.18	1.42	8.35	21.53	1.37	49.65	71.56	56.82	68.10	45.76	22.34	19.25	**36.39**
HIMDLA (2024) [16]	6.72	50.16	28.72	37.93	34.02	46.15	2.97	1.33	7.82	22.15	1.28	46.47	66.96	52.18	63.73	42.82	20.91	20.02	**34.05**
STSNet (2024) [17]	6.09	54.35	28.41	40.34	36.18	52.27	3.16	1.41	8.32	21.43	1.36	49.42	73.22	56.55	68.78	45.54	22.23	19.16	**36.21**
PILFR (2024) [18]	6.94	52.05	27.73	39.36	35.31	54.00	3.09	1.38	8.11	20.91	1.33	48.22	69.49	57.18	66.14	44.44	22.70	18.70	**35.34**
Ours	6.24	53.68	29.12	44.34	36.09	53.58	3.24	1.45	7.52	21.97	1.40	52.66	72.00	56.97	70.47	47.68	22.79	19.64	**37.12**

generates attention maps that cover diverse object regions without spatial bias, whereas Gaussian sampling overfocuses on central areas and uniform sampling lacks focus, which demonstrates that random sampling optimally balances exploration and exploitation for fine-grained feature learning. In general, our counterbalancing mechanism with randomly generating fake attention maps suppresses disruptive background activations while adaptively recalibrating feature significance. It effectively mitigates model bias toward visually obvious patterns that frequently cause misclassification in fine-grained tasks.

2) Loss Function: Table II shows the ablation study of the loss function. With the addition of the loss function $L(Y, S_c)$, performance increases to 32.53% mAP, and 60.35% CorLoc on the FAIR1M-v1.0 [26], and 49.11% mAP and 68.32% CorLoc on the ShipRSImageNet [55]. Adding the function L_r^k further boosts mAP and CorLoc to 34.97% mAP and 64.87% CorLoc on the FAIR1M-v1.0 [26], and 52.79% mAP and 73.44% CorLoc on the ShipRSImageNet [55]. Finally, including the L_{reg} results in the highest performance for 37.12% mAP and 68.86% CorLoc on the FAIR1M-v1.0 [26] and 56.04% mAP and 77.96% CorLoc on the ShipRSImageNet [55]. Each additional loss function improves the model's performance on both datasets.

In terms of inference time, while there is a slight increase as more loss functions are added, the increase is minimal to 3.2 ms approximately, which is acceptable considering the significant performance improvements. While the number of parameters remains constant (195M), the FLOPs decrease slightly with each added loss function, suggesting that the additional loss functions lead to better computational efficiency at the expense of only a small increase in parameters. The tradeoff between slightly higher inference time and improved performance (particularly in FGOD) is acceptable in this context.

3) Multiorder Design in LMD: The second-order (2-order in Table III) degradation process is selected to balance computational efficiency and real-world applicability. Table III shows the ablation study of the loss function. The 2-order degradation shows further improvement, with 37.12% mAP and 68.86% CorLoc on FAIR1M-v1.0 [26], and 56.04% mAP and 77.96% CorLoc on ShipRSImageNet [55]. The 3-order degradation achieves the best results with 37.84% mAP and 70.20% CorLoc on FAIR1M-v1.0 [26], and 57.13% mAP and 79.47% CorLoc on ShipRSImageNet [55]. Increasing the order of LMD improves the performance on both datasets significantly, especially in terms of mAP and CorLoc.

In terms of computational complexity, Table III reveals critical tradeoffs in the multiorder design of LMD. Inference time remains relatively stable across different degradation orders, highlighting the lightweight nature of the design. Adding more orders of the degradation in LMD implies more parameters and lower FLOPs. For instance, with the VGG16 backbone, parameters grow from 145M of 1-order to 195M of 2-order and 268M of 3-order, indicating that the increase in parameters with higher-order LMD is substantial. However, FLOPs show a nonmonotonic trend, decreasing from 142.07G of 1-order to 132.02G of 2-order before slightly rising to 134.56G of 3-order for VGG16, suggesting optimized computation in the 2-order degradation. This change is more noticeable when using more complex backbones such as ResNet-50-FPN and LSKNet, but it is also accompanied by better performance

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TABLE V

CORRECT LOCALIZATION (%) RESULTS FOR DIFFERENT WSOD METHODS ON THE FAIR1M-V1.0 AND SHIPRSIMAGENET TRAIN–VAL DATASETS

WSOD Methods	WSDDN [5]	OICR [6]	PCL [7]	PCIR [3]	DCL [1]	TCANet [10]	SAENet [11]
FAIR1M-v1.0	52.30	56.38	57.76	59.77	57.65	61.34	64.04
ShipRSImageNet	56.04	62.33	64.00	67.31	68.79	68.30	71.13
WSOD Methods	RINet [4]	WSONet [14]	PSL [15]	HIMDLA [16]	STSNet [17]	PILFR [18]	Ours
FAIR1M-v1.0	65.07	64.14	67.26	66.73	68.07	67.68	68.86
ShipRSImageNet	72.81	71.56	75.96	74.02	76.00	75.85	77.96

TABLE VI

AVERAGE PRECISION (%) RESULTS FOR DIFFERENT WSOD METHODS ON THE SHIPRSIMAGENET TESTING DATASET

WSOD Methods	OS	OW	Su	OAC	En	Ni	Mi	Ti	OD	AtD	ABD	HaD	HyD	AsD	OF	PF	Pa
WSDDN (2016) [5]	22.30	27.46	47.00	49.03	48.56	45.35	49.78	48.98	33.11	44.86	49.17	48.26	49.67	45.17	36.82	48.79	27.01
OICR (2017) [6]	24.80	30.55	52.28	54.89	54.01	50.45	55.89	54.48	36.83	49.90	54.69	53.68	55.25	50.24	40.96	54.27	30.05
PCL (2018) [7]	25.47	31.36	53.68	56.64	55.46	51.80	57.64	55.94	37.81	51.23	56.15	55.12	56.73	51.59	42.06	55.72	30.85
PCIR (2020) [3]	26.36	32.46	55.56	58.89	57.40	53.62	59.89	57.90	39.14	53.03	58.12	57.05	58.72	53.40	43.53	57.68	31.93
DCL (2021) [1]	27.38	33.71	57.70	61.67	59.61	55.68	62.42	60.13	40.65	55.07	60.36	59.25	60.98	55.45	45.21	59.90	33.17
TCANet (2021) [10]	27.18	33.47	57.29	61.48	59.19	55.29	59.98	59.71	40.36	54.68	59.93	58.83	60.54	55.06	44.89	59.47	32.93
SAENet (2022) [11]	28.48	35.07	60.02	64.64	62.01	57.92	63.14	62.55	42.28	57.29	62.79	61.63	63.43	57.68	47.03	62.31	34.50
RINet (2022) [4]	30.02	36.97	63.27	66.34	65.37	61.06	67.09	65.94	44.57	60.39	66.19	64.97	66.86	60.81	49.57	65.68	36.37
WSONet (2023) [14]	28.97	35.68	61.07	66.00	63.09	58.93	64.50	63.64	43.02	58.29	63.88	62.71	64.54	58.69	47.85	63.39	35.10
PSL (2023) [15]	30.40	37.43	64.07	67.44	66.19	61.83	68.44	66.77	45.13	61.15	67.02	65.79	67.71	61.57	50.20	66.51	36.82
HIMDLA (2024) [16]	28.30	34.86	59.66	63.00	61.64	57.57	64.00	62.17	42.02	56.94	62.41	61.26	63.05	57.33	46.74	61.93	34.29
STSNet (2024) [17]	30.24	37.24	63.74	67.60	65.86	61.51	68.60	66.43	44.90	60.84	66.68	65.45	67.36	61.26	49.94	66.17	36.64
PILFR (2024) [18]	29.46	36.28	62.09	66.33	64.14	59.91	67.08	64.70	43.74	59.26	64.95	63.75	65.61	59.67	48.64	64.45	35.69
Ours	30.05	37.24	67.45	71.67	65.62	61.16	67.17	71.21	45.11	64.47	71.47	69.21	72.17	64.90	49.28	65.95	36.62

WSOD Methods	OL	YTL	YDeL	YDaL	YZL	AL	OsL	WaL	LSD	LHA	Co	OAS	MS	TeS	TrS	AOE	MAS
WSDDN (2016) [5]	26.44	47.01	47.16	43.13	50.78	47.87	49.53	50.28	44.03	48.58	49.84	26.07	41.89	45.56	51.03	49.28	50.53
OICR (2017) [6]	29.41	52.29	52.46	47.97	56.64	53.25	55.39	56.14	48.97	54.04	55.44	29.00	46.59	50.68	54.64	55.14	56.39
PCL (2018) [7]	30.20	53.69	53.86	49.26	56.14	54.68	57.39	57.89	50.29	55.49	56.92	29.77	47.84	52.04	56.39	56.89	58.14
PCIR (2020) [3]	31.26	55.57	55.75	50.98	58.39	56.59	59.64	60.14	52.05	57.43	58.92	30.82	49.52	53.86	58.64	59.39	58.14
DCL (2021) [1]	32.46	57.71	57.90	52.95	60.92	58.78	62.17	60.42	54.05	59.65	61.19	32.00	51.42	55.94	61.17	61.92	60.67
TCANet (2021) [10]	32.23	57.30	57.49	52.57	60.73	58.36	61.98	60.23	53.67	59.22	60.75	31.78	51.06	55.54	61.23	61.73	60.48
SAENet (2022) [11]	33.77	60.03	60.23	55.08	64.14	61.14	62.89	63.39	56.23	62.04	63.65	33.29	53.49	58.18	64.39	64.89	63.64
RINet (2022) [4]	35.59	63.28	63.49	58.06	68.09	64.45	66.84	67.59	59.27	65.40	67.09	35.09	56.39	61.33	68.34	66.59	67.84
WSONet (2023) [14]	34.36	61.08	61.28	56.04	65.50	62.20	64.25	64.75	57.21	63.13	64.76	33.87	54.42	59.20	65.75	64.00	65.25
PSL (2023) [15]	36.04	64.08	64.29	58.79	69.19	65.26	67.94	68.69	60.02	66.23	67.94	35.54	57.10	62.11	67.19	67.69	68.94
HIMDLA (2024) [16]	33.56	59.67	59.86	54.74	62.50	60.77	63.75	64.25	55.89	61.67	63.26	33.09	53.17	57.83	62.75	63.25	64.50
STSNet (2024) [17]	35.86	63.76	63.96	58.49	67.10	64.93	68.35	68.85	59.71	65.89	67.60	35.36	56.81	61.79	67.35	68.10	66.85
PILFR (2024) [18]	34.93	62.10	62.30	56.97	65.58	63.24	66.83	65.08	58.16	64.18	65.84	34.44	55.33	60.19	65.83	66.58	65.33
Ours	35.82	67.47	66.68	62.06	67.92	68.67	68.17	67.42	63.31	69.66	72.41	35.30	60.33	65.45	67.42	67.92	67.67

WSOD Methods	SAS	EPF	OM	CS	Ro	Ca	Ba	Tu	Fe	Ya	Sa	FV	OT	Ho	Mo	Do	**mAP**
WSDDN (2016) [5]	47.66	46.77	10.07	37.48	45.08	40.77	22.20	36.01	26.47	44.27	20.74	28.06	38.32	24.90	14.66	38.26	**40.24**
OICR (2017) [6]	53.01	52.03	11.20	41.69	50.14	45.35	24.70	40.06	29.44	49.25	23.07	31.21	42.62	27.69	16.30	42.56	**44.76**
PCL (2018) [7]	54.43	53.42	11.50	42.81	51.49	46.56	25.36	41.13	30.23	50.57	23.69	32.05	43.76	28.44	16.74	43.70	**45.96**
PCIR (2020) [3]	56.34	55.29	11.90	44.31	53.29	48.20	26.25	42.57	31.29	52.34	24.52	33.17	45.30	29.43	17.33	45.23	**47.57**
DCL (2021) [1]	58.51	57.42	12.36	46.02	55.34	50.05	27.26	44.21	32.50	54.36	25.46	34.45	47.04	30.57	17.99	46.97	**49.40**
TCANet (2021) [10]	58.09	57.01	12.28	45.69	54.95	49.70	27.06	43.90	32.27	53.97	25.28	34.20	46.71	30.35	17.87	46.64	**49.05**
SAENet (2022) [11]	60.86	59.73	12.86	47.87	57.57	52.07	28.35	45.99	33.81	56.54	26.49	35.83	48.93	31.80	18.72	48.86	**51.39**
RINet (2022) [4]	64.16	62.97	13.56	50.46	60.68	54.88	29.89	48.48	35.64	59.60	27.92	37.77	51.58	33.52	19.73	51.51	**54.17**
WSONet (2023) [14]	61.92	60.77	13.09	48.70	58.57	52.97	28.85	46.79	34.39	57.53	26.95	36.46	49.79	32.35	19.04	49.71	**52.29**
PSL (2023) [15]	64.97	63.76	13.73	51.09	61.45	55.58	30.27	49.09	36.08	60.35	28.27	38.25	52.23	33.94	19.98	52.16	**54.85**
HIMDLA (2024) [16]	60.49	59.37	12.78	47.58	57.22	51.75	28.18	45.71	33.60	56.20	26.32	35.62	48.64	31.60	18.60	48.57	**51.08**
STSNet (2024) [17]	64.64	63.44	13.66	50.83	61.14	55.29	30.11	48.84	35.90	60.05	28.13	38.06	51.97	33.77	19.88	51.89	**54.58**
PILFR (2024) [18]	62.96	61.79	13.30	49.51	59.55	53.86	29.33	47.57	34.97	58.49	27.40	37.07	50.62	32.89	19.36	50.54	**53.16**
Ours	68.37	63.14	13.02	52.20	60.78	54.78	29.92	50.15	35.86	63.66	27.88	38.08	53.36	33.67	19.41	53.28	**56.04**

in terms of mAP and CorLoc. While the 3-order degradation, respectively, improves mAP by only 0.72% and 1.09% over the 2-order model on the FAIR1M-v1.0 [26] and ShipRSImageNet [55] datasets, it incurs a 35% higher computational cost. Considering the potential impact on practical applications, this diminishing return suggests that the 2-order degradation achieves an optimal tradeoff between accuracy and efficiency for practical deployments, while the 3-order offers only little improvements at a higher computational cost. Therefore, it is recommended to leverage lightweight backbones and optimize the 2-order degradation pipeline through modular feature reuse.

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TABLE VII

AVERAGE PRECISION (%) RESULTS FOR DIFFERENT FGOD METHODS ON THE SHIPRSIMAGENET TESTING DATASET. (HERE, ALL OTHER METHODS EXCEPT OURS ARE FULLY SUPERVISED; R-50 REFERS TO RESNET-50)

FGOD Methods	One-stage				Two-stage				Ours	
	FCOS [51]			S2ANet [52]	RoI Trans [53]		PETDet [54]			
Backbone	R-50	R-50 w/OM	R-50-FPN	R-50-FPN	R-50	R-50-FPN	R-50 w/OM	R-50-FPN	VGG16	R-50
mAP of ShipRSImageNet	26.58	30.66	31.18	36.02	29.84	33.56	48.99	55.69	56.04	57.45

TABLE VIII

AVERAGE PRECISION (%) RESULTS FOR DIFFERENT FGOD METHODS ON THE FAIR1M-V1.0 TESTING DATASET. (HERE, ALL OTHER METHODS EXCEPT OURS ARE FULLY SUPERVISED; R-50 REFERS TO RESNET-50)

FGOD Methods	**Backbone**	B737	B747	B777	B787	C919	A220	A321	A330	A350	ARJ21	OA	PS	MB	FB	TB	ES	LCS	DCS	WS
FCOS (2019) [51]	R-50	36.95	79.70	12.15	36.76	1.48	48.48	66.04	61.19	70.62	4.14	-	8.54	52.52	8.57	30.52	11.87	13.82	37.39	23.00
FCOS (2019) [51]	R-50 w/OM	35.13	81.78	12.89	39.74	11.64	46.72	66.52	61.19	70.14	26.84	-	8.95	53.66	8.82	33.98	10.93	13.70	36.52	22.21
FCOS (2019) [51]	R-50	38.12	89.30	31.65	54.72	0.62	52.88	69.63	50.31	58.98	3.15	-	11.28	54.12	16.19	29.83	13.97	46.74	49.55	28.82
S2ANet (2021) [52]	R-50	36.06	84.33	15.62	42.34	1.95	44.09	68.00	63.84	70.00	12.10	-	8.82	48.03	6.79	34.01	7.49	18.30	37.62	22.96
S2ANet (2021) [52]	R-50-FPN	39.74	88.74	33.51	52.09	0.50	49.34	70.52	47.54	47.07	2.20	-	4.80	23.47	4.67	15.62	7.59	31.81	33.34	7.16
RoI Trans (2019) [53]	R-50	40.14	84.92	15.39	49.17	19.73	50.46	70.31	71.42	72.62	33.65	-	13.21	56.54	6.82	35.71	9.96	16.91	36.01	17.30
RoI Trans (2019) [53]	R-50-FPN	46.52	93.05	41.60	62.88	9.90	53.80	71.07	61.48	67.02	14.35	-	12.71	58.74	19.33	30.79	13.81	46.96	50.52	27.33
PCLDet (2023) [42]	R-50	43.51	87.37	28.73	55.37	20.04	51.65	73.06	74.52	79.86	31.08	-	18.21	62.88	12.05	35.17	13.52	23.79	42.46	37.45
SFRNet (2023) [33]	R-50-FPN	48.26	86.54	43.16	65.24	10.27	55.82	73.73	63.78	69.53	24.89	-	13.19	60.94	20.05	31.94	14.33	48.72	52.41	28.35
RB-FPN (2023) [43]	R-50	39.16	86.70	16.17	49.16	13.60	49.18	66.91	71.43	74.34	29.00	-	18.38	68.41	10.55	38.27	11.83	25.12	38.41	34.72
PETDet (2024) [54]	R-50	41.05	82.23	22.04	51.21	25.66	51.83	69.53	70.86	73.70	36.90	-	13.29	61.94	8.90	37.88	10.93	22.05	36.82	23.98
PETDet (2024) [54]	R-50 w/OM	42.38	82.70	20.07	55.71	22.91	53.61	70.11	71.16	71.46	30.60	-	14.56	63.36	9.30	37.63	11.64	23.36	36.82	24.05
QEDetr (2025) [41]	R-50	49.02	94.36	36.91	64.33	26.44	52.90	71.79	60.30	73.94	22.76	-	16.10	70.13	28.63	31.22	22.69	55.06	55.06	43.80
QEDetr (2025) [41]	Swin-t	49.71	94.38	41.67	65.07	26.92	55.46	72.40	67.93	76.27	20.24	-	17.20	73.31	33.74	32.48	23.94	56.01	55.67	45.69
Ours	VGG16	64.52	53.59	35.64	37.30	27.93	64.00	69.05	46.35	31.15	32.47	65.37	7.16	36.63	19.93	39.46	38.37	47.06	51.06	7.58
Ours	R-50	66.16	54.95	36.54	38.25	28.64	65.63	70.80	47.52	31.94	33.29	67.03	7.34	37.56	20.44	40.46	39.35	48.25	52.35	7.77

FGOD Methods	**Backbone**	OS	SC	BUS	CT	DT	VAN	TRAI	TRAC	EX	TT	OV	BC	TC	FF	BF	IS	RO	BR	**mAP**
FCOS (2019) [51]	R-50	-	52.22	22.00	33.20	28.00	45.93	11.11	2.45	11.21	0.44	-	40.97	78.41	53.69	87.56	55.79	17.39	15.08	**34.08**
FCOS (2019) [51]	R-50 w/OM	-	52.13	21.51	34.44	28.91	46.94	9.73	1.21	12.12	0.53	-	42.18	78.17	57.24	86.93	56.13	19.20	12.53	**35.33**
FCOS (2019) [51]	R-50	-	54.92	12.40	44.78	34.71	51.71	12.19	1.24	13.39	22.79	-	50.64	86.89	58.43	89.04	65.88	27.51	32.89	**39.98**
S2ANet (2021) [52]	R-50	-	61.57	11.76	34.28	36.03	54.62	3.47	0.96	7.24	0.40	-	38.44	80.44	56.34	87.47	50.76	16.67	17.24	**34.71**
S2ANet (2021) [52]	R-50-FPN	-	47.77	2.38	19.75	24.10	39.53	0.08	0.08	1.71	0.02	-	40.69	79.13	46.33	87.14	56.07	29.15	18.02	**30.93**
RoI Trans (2019) [53]	R-50	-	58.29	28.02	40.55	45.97	54.10	11.82	2.61	11.74	0.72	-	47.50	80.12	58.03	87.37	58.58	21.85	23.52	**39.15**
RoI Trans (2019) [53]	R-50-FPN	-	58.10	31.76	49.76	48.78	52.92	13.68	1.81	13.21	16.34	-	56.90	88.88	62.80	90.81	64.40	31.90	32.99	**44.03**
PCLDet (2023) [42]	R-50	-	61.26	29.13	42.81	47.99	56.93	14.57	5.19	13.08	1.36	-	55.01	78.69	69.55	90.69	62.35	20.55	39.30	**43.50**
SFRNet (2023) [33]	R-50-FPN	-	60.28	32.95	51.62	50.61	54.90	14.19	1.88	13.71	16.95	-	59.03	92.21	65.15	94.21	66.81	33.10	34.23	**45.68**
RB-FPN (2023) [43]	R-50	-	70.75	36.14	44.94	50.20	70.77	16.75	1.68	17.24	0.49	-	54.59	80.49	65.69	88.92	56.64	20.36	32.19	**42.62**
PETDet (2024) [54]	R-50	-	68.81	18.33	42.17	47.15	65.48	11.93	2.09	8.51	0.41	-	42.64	78.60	63.51	87.40	57.48	20.82	25.57	**40.55**
PETDet (2024) [54]	R-50 w/OM	-	68.64	30.02	42.76	47.63	65.36	10.73	2.08	10.58	0.55	-	45.35	79.54	63.10	87.21	57.11	24.86	23.44	**41.19**
QEDetr (2025) [41]	R-50	-	74.20	39.27	56.75	57.17	73.56	16.28	2.63	18.09	37.54	-	57.36	89.64	59.18	88.72	54.15	21.40	30.55	**48.59**
QEDetr (2025) [41]	Swin-t	-	75.08	38.76	57.46	58.17	74.22	16.06	2.27	19.64	43.08	-	57.85	90.97	54.51	87.58	55.09	16.62	35.53	**49.74**
Ours	VGG16	6.24	54.68	29.12	41.34	37.09	53.58	3.24	1.45	8.52	21.97	1.40	50.66	73.00	57.97	69.47	46.68	22.79	19.64	**37.12**
Ours	R-50	6.40	56.06	29.86	42.39	38.03	54.93	3.32	1.48	8.74	22.52	1.43	51.94	74.85	59.43	71.23	47.86	23.37	20.14	**38.06**

In conclusion, the higher-order degradation, such as 3-order, could simulate more complex scenarios and achieve better performance, but it significantly increases model complexity and training time without proportional performance gains.

D. Comparative Experiments With State-of-the-Arts

1) Comparison With SOTA WSOD Methods: To verify the effectiveness of EFDNet, we conduct extensive comparisons on the FAIR1M-v1.0 [26] and ShipRSImageNet [55] datasets with 13 existing SOTA WSOD methods. Considering that some WSOD methods like RINet [4], PSL [15], HIMDLA [16], STSNet [17], and PILFR [18] have not been tested on both datasets, we keep the same training settings as [6], [7], and [8] mentioned above to test them.

Tables IV and V, respectively, show quantitative comparisons of mAP and CorLoc for each category with 13 existing SOTA WSOD methods on the FAIR1M-v1.0 [26] dataset, which shows that our EFDNet, respectively, achieves 37.12% mAP and 68.86% CorLoc. Since we have selected these advanced WSOD methods, our EFDNet outperforms these approaches in about half of the 37 subcategories and gets the second best in about half of the categories, while the remaining quarter of the categories are also very close to the SOTA methods. Here, the best and second-best indicators are, respectively, highlighted in red and blue. Specifically, thanks to the high-quality proposals generated by ACPR, take 11 subcategories of airplane [i.e., B737, B747, B777, B787, C919, A220, A321, A330, A350, ARJ21, and other-airplane (OA)] on the FAIR1M-v1.0 [26] dataset as an example, our network performs the best in more than half of these subcategories in the detection of tiny objects overall.

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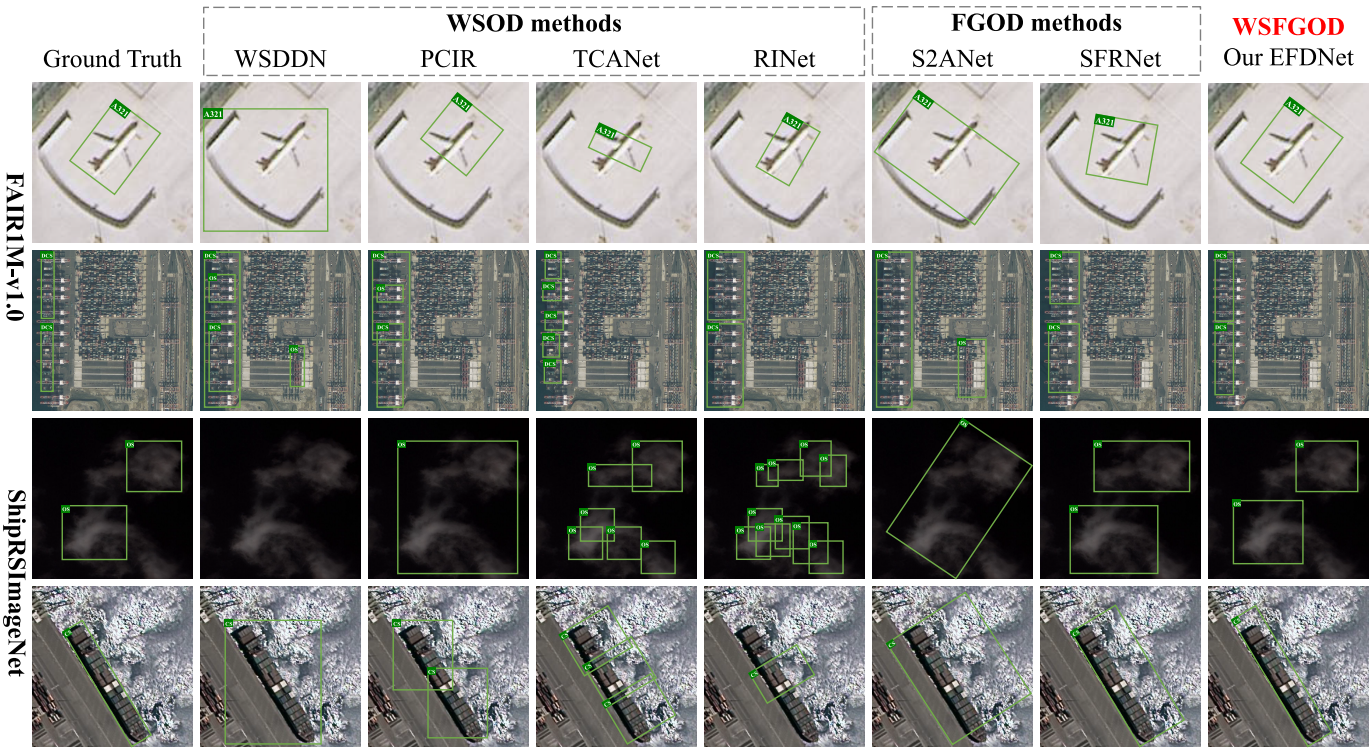


Fig. 6. Comparison of four WSOD and two FGOD advanced methods with our EFDNet on the FAIR1M-v1.0 and ShipRSImageNet datasets.



Fig. 7. More qualitative detection results on the FAIR1M-v1.0 and ShipRSImageNet datasets.

As for the ShipRSImageNet [55] dataset, Tables V and VI, respectively, show quantitative comparisons of CorLoc and mAP for each category with existing advanced SOTA WSOD methods, which show that our EFDNet, respectively, achieves 56.04% mAP and 77.96% CorLoc. According to comparative experiments on the ShipRSImageNet [55] dataset, our EFDNet outperforms these advanced WSOD approaches in nearly half of the 50 categories and gets the second best in 6 categories, while the remaining two-fifths of the categories are also very close to the SOTA methods.

2) Comparison With SOTA FGOD Methods: Almost all existing FGOD methods are based on the fully supervised learning framework (see Tables VII and VIII). As we know, the performance of object detection methods based on the fully supervised learning framework is often better than that of the weakly supervised learning framework under the same conditions of training model and backbone network.

Tables VII and VIII show quantitative comparisons of mAP for each category with existing advanced FGOD methods on the FAIR1M-v1.0 [26] and ShipRSImageNet [55] datasets. For the ResNet-50 backbone, our weakly supervised EFDNet achieves 38.06% and 57.45% mAP on the two datasets, which is within −11.68% and +1.76% of the fully supervised SOTA, demonstrating the viability of the WSFGOD paradigm. For the price of only image-level labels, EFDNet can close the performance gap to fully supervised methods to just 4.96% on the two datasets, representing a massive leap in annotation efficiency for fine-grained tasks. Noting that most existing FGOD methods are based on the fully supervised learning framework and they do not consider three subcategories [i.e., OA, OS, and other-vehicle (OV)] when calculating the mAP, we compare 37 subcategories of our EFDNet with these FGOD methods based on the fully supervised learning framework, rather than the total mAP of the FAIR1M-v1.0 [26] dataset. Obviously, our EFDNet outperforms these approaches in 5 of 37 subcategories, which represents that our method completely eliminates the need for expensive and fine-grained bounding box annotations.

In conclusion, comprehensive comparison experiments show that the proposed EFDNet significantly surpasses exist-

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ing SOTA WSOD methods and is close to existing fully supervised FGOD methods, which demonstrates its robustness.

E. Visualization Analysis

Fig. 6 illustrates the visual comparison results obtained from the FAIR1M-v1.0 [26] and ShipRSImageNet [55] datasets, where objects are delineated with green bounding boxes. Additionally, the category of each object is labeled alongside the bounding box. Compared with four SOTA WSOD methods [3], [4], [5], [10] and two SOTA FGOD methods [33], [52], our approach achieves more precise localization and more accurate identification for small objects in multiscale problems. Take A321, dry cargo ship (DCS), and other-ship (OS) from FAIR1M-v1.0 [26] and container ship (CS) from ShipRSImageNet [55] in Fig. 6, and tennis court (TC), ARJ21 in Fig. 7 as an example, the proposals are detected well when their object rotation angle is small and the gap with the background features is large. Comprehensive visualization experiments show that the proposed EFDNet effectively alleviates the issues of top-scoring proposals and RSI interference.

V. Conclusion

In this article, we propose a novel EFDNet to perform WSFGOD in RSIs. We first design a novel LMD module for real-world degradation to obtain higher-resolution image features. Based on the WSOD framework, ACPR is designed to divert the focus of the detection network from the local distinct part to the object and further to other potential instances by leveraging both local and global context information. The FDH module utilizes CB to extract discriminative contextual semantic features from the input images by introducing fake attention maps, while LB further extracts edge semantic information. On the challenging FAIR1M-v1.0 and ShipRSImageNet datasets, extensive experiments clearly demonstrate that the proposed method is quite effective in addressing the multiscale issue and achieves state-of-the-art performance, which means that the proposed method significantly boosts object detection accuracy compared with most existing weakly supervised and fine-grained approaches.

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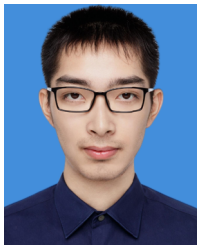
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